

MATH 15200: Calculus II

Homework 1: Limits and Continuity

Due: Monday, October 5, 2026, at 11:59 PM.

This is the written portion of Homework 1; complete the routine online portion separately in Edfinity. Upload your written solutions to Canvas. You may collaborate, but you must write and submit your own solutions independently.

Definition. Let f be a real-valued function defined near a , possibly not at a . We say

$$\lim_{x \rightarrow a} f(x) = L$$

if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that, whenever x is in the domain of f ,

$$0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon.$$

If $f(a)$ is defined, then f is **continuous at** a when

$$\lim_{x \rightarrow a} f(x) = f(a).$$

Rewrite each of the problems below in terms of the epsilon-delta definition above. Explicitly state your choice of δ when necessary.

1. Prove directly from the epsilon-delta definition that

$$\lim_{x \rightarrow 2} (x^2 - 1) = 3.$$

Hint: Impose a simple bound such as $|x - 2| < 1$.

2. Define a function f by the piecewise expression

$$f(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0. \end{cases}$$

Prove directly from the definition that f is not continuous at 0.

Hint: Choose a fixed positive ε and consider points immediately to the left of 0.

3. Suppose that f and g are real-valued functions defined near a and that

$$\lim_{x \rightarrow a} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = M.$$

Prove from the epsilon-delta definition that

$$\lim_{x \rightarrow a} (f(x) + g(x)) = L + M.$$

Hint: Apply the two hypotheses using $\varepsilon/2$ and combine the resulting choices of δ .